

Movie Studio Investment Strategy: An Exploratory Data Analysis

Author: Kevin **Contact:** [LinkedIn](#) | kevin@example.com

This notebook analyzes box office, budget, studio, and audience-rating data to answer three questions for our company's new movie studio leadership:

1. **Which genres deliver the best financial return** (measured by ROI and median profit)?
2. **Which genres do best with an international audience?**
3. **Which studios perform best, and do they align with the top genres?**

We use `pandas` to combine data from **The Numbers** (production budgets and worldwide grosses), **Box Office Mojo** (studio and foreign/domestic gross split), and **IMDB** (genres and audience ratings) into a single analytical dataset of over 1,000 major theatrical releases.

Table of Contents

1. [Business Understanding](#)
2. [Data Understanding](#)
3. [Data Preparation & Cleaning](#)
4. [Data Analysis & Visualization](#)
 - [Question 1: Which genres deliver the best financial return?](#)
 - [Question 2: Which genres do best with an international audience?](#)
 - [Question 3: Which studios perform best, and do they align with the top genres?](#)
5. [Conclusions & Recommendations](#)

Business Understanding

Stakeholder: The head of our company's brand-new movie studio division.

Real-world problem: Our company has decided to enter the movie business but has no track record, no established franchises, and no in-house filmmaking expertise yet. Before the studio commits capital to any project, its leadership needs an evidence-based starting point for what kinds of films to greenlight - which genres pay off, which genres travel well internationally, and which existing studios are worth benchmarking against.

This notebook directly supports that decision by answering three specific business questions:

1. Which genres deliver the best financial return, measured by ROI and median profit?
2. Which genres do best with an international audience?
3. Which studios perform best, and do their genre choices align with the top-performing genres?

```
In [1]: # Imports
import sqlite3

import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import numpy as np
import pandas as pd
import seaborn as sns

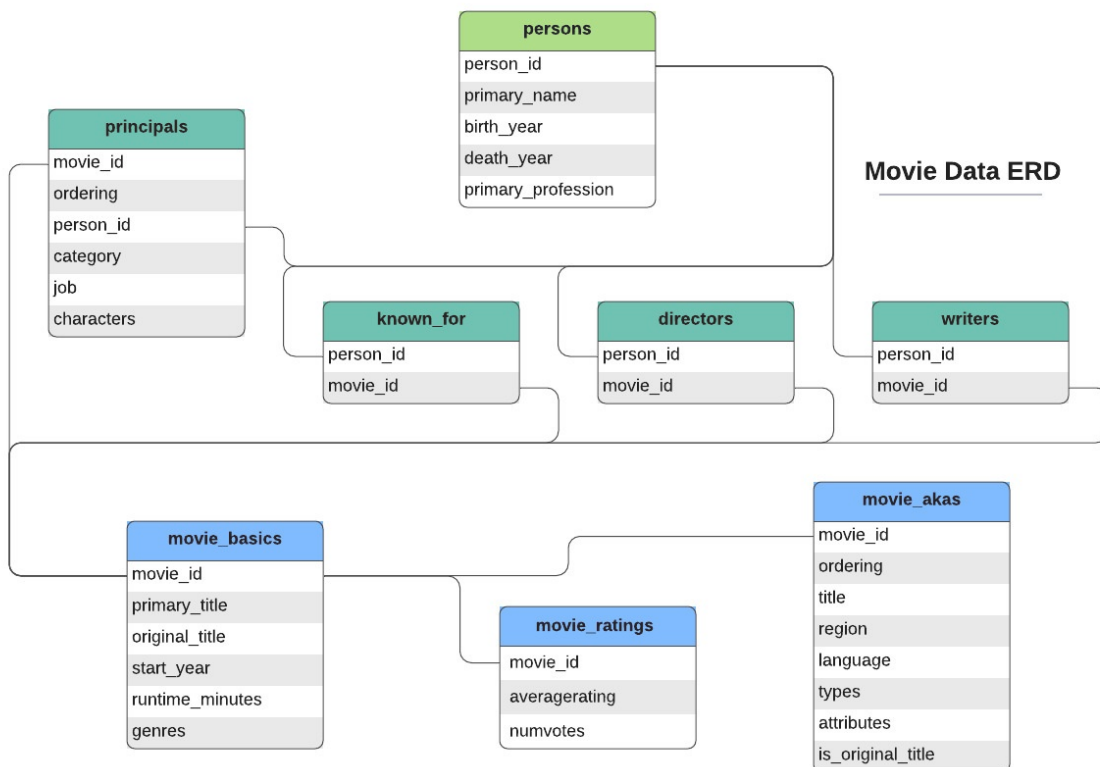
sns.set_theme(style="whitegrid", context="talk")
PALETTE = ["#1E3D59", "#FF6E40", "#FFC13B", "#3FA796", "#A6A6A6"]
pd.set_option("display.max_columns", 20)
%matplotlib inline
```

Data Understanding

We draw on three of the provided data sources:

Source	File	What it provides
The Numbers	zippedData/tn.movie_budgets.csv.gz	Production budget, domestic gross, worldwide gross, release date
Box Office Mojo	zippedData/bom.movie_gross.csv.gz	Studio, domestic gross, foreign gross
IMDB	zippedData/im.db (movie_basics , movie_ratings tables)	Genre(s), runtime, average user rating, number of votes

The entity-relationship diagram below shows how the IMDB tables relate to each other. We only use the `movie_basics` and `movie_ratings` tables (highlighted in blue), joined on `movie_id`.



Rotten Tomatoes and TheMovieDB data are included in `zippedData/` but are not used here: neither shares a title/year combination reliable enough to merge cleanly with the other three sources (a documented limitation, see Conclusions).

Let's load and inspect each source before cleaning anything.

```
In [2]: # Load The Numbers budgets data
tn_raw = pd.read_csv("zippedData/tn.movie_budgets.csv.gz")
print(tn_raw.shape)
tn_raw.head()
```

(5782, 6)

```
Out[2]:
```

	id	release_date	movie	production_budget	domestic_gross	worldwide_gross
0	1	Dec 18, 2009	Avatar	\$425,000,000	\$760,507,625	\$2,776,345,279
1	2	May 20, 2011	Pirates of the Caribbean: On Stranger Tides	\$410,600,000	\$241,063,875	\$1,045,663,875
2	3	Jun 7, 2019	Dark Phoenix	\$350,000,000	\$42,762,350	\$149,762,350
3	4	May 1, 2015	Avengers: Age of Ultron	\$330,600,000	\$459,005,868	\$1,403,013,963
4	5	Dec 15, 2017	Star Wars Ep. VIII: The Last Jedi	\$317,000,000	\$620,181,382	\$1,316,721,747

```
In [3]: tn_raw.dtypes
```

```
Out[3]: id                int64
release_date            str
movie                   str
production_budget       str
domestic_gross          str
worldwide_gross         str
dtype: object
```

```
In [4]: # Load Box Office Mojo data
bom_raw = pd.read_csv("zippedData/bom.movie_gross.csv.gz")
print(bom_raw.shape)
bom_raw.head()
```

(3387, 5)

```
Out[4]:
```

	title	studio	domestic_gross	foreign_gross	year
0	Toy Story 3	BV	415000000.0	652000000	2010
1	Alice in Wonderland (2010)	BV	334200000.0	691300000	2010
2	Harry Potter and the Deathly Hallows Part 1	WB	296000000.0	664300000	2010
3	Inception	WB	292600000.0	535700000	2010
4	Shrek Forever After	P/DW	238700000.0	513900000	2010

```
In [5]: bom_raw.dtypes
```

```
Out[5]: title                str
studio                   str
domestic_gross          float64
foreign_gross           str
year                    int64
dtype: object
```

```
In [6]: # Load the IMDB SQLite database
conn = sqlite3.connect("zippedData/im.db")

basics_raw = pd.read_sql(
    "SELECT movie_id, primary_title, start_year, runtime_minutes, genres FROM movie_basics",
    conn,
)
ratings_raw = pd.read_sql(
    "SELECT movie_id, averagerating, numvotes FROM movie_ratings", conn
)
conn.close()

print("movie_basics:", basics_raw.shape)
print("movie_ratings:", ratings_raw.shape)
basics_raw.head()
```

movie_basics: (146144, 5)

movie_ratings: (73856, 3)

Out [6]:	movie_id	primary_title	start_year	runtime_minutes	genres
0	tt0063540	Sunghursh	2013	175.0	Action,Crime,Drama
1	tt0066787	One Day Before the Rainy Season	2019	114.0	Biography,Drama
2	tt0069049	The Other Side of the Wind	2018	122.0	Drama
3	tt0069204	Sabse Bada Sukh	2018	NaN	Comedy,Drama
4	tt0100275	The Wandering Soap Opera	2017	80.0	Comedy,Drama,Fantasy

Data understanding takeaways:

- `tn.movie_budgets` has currency figures stored as strings (e.g. "\$425,000,000") - these need cleaning before they can be used numerically.
- `bom.movie_gross` has `domestic_gross` as a float already, but `foreign_gross` is stored as a string and is missing for a large share of rows (checked below).
- `movie_basics` has far more titles than just theatrical releases (shorts, TV movies, straight-to-video, etc.), and only titles with a matching row in `movie_ratings` have received enough audience votes to be meaningfully rated.
- `genres` is a comma-separated string (e.g. "Action,Adventure,Sci-Fi"), so a film can belong to multiple genres.
- None of the three sources share a single common ID - they must be joined on matching **movie title + release year**.

Data Preparation & Cleaning

To answer all three business questions we need one row per movie with financial data (budget, domestic gross, foreign gross), genre, and studio. We build this via the following reproducible, documented cleaning steps.

Step 1: Clean The Numbers (`tn`)

- Strip \$ and , from `production_budget`, `domestic_gross`, `worldwide_gross` and cast to `float`.
- Parse `release_date` into a proper datetime and extract `year`.
- Drop rows with a \$0 production budget - these are missing-data placeholders, not real free films, and would break ROI calculations (division by zero).

```
In [7]: tn = tn_raw.copy()

for col in ["production_budget", "domestic_gross", "worldwide_gross"]:
    tn[col] = tn[col].str.replace(r"[\$,]", "", regex=True).astype(float)

tn["release_date"] = pd.to_datetime(tn["release_date"])
tn["year"] = tn["release_date"].dt.year
tn = tn.rename(columns={"movie": "title"})

before = len(tn)
tn = tn[tn["production_budget"] > 0]
print(f"Dropped {before - len(tn)} rows with a $0 placeholder budget")
print(f"Duplicate titles remaining (legitimate remakes/re-releases): {tn['title'].duplicated().sum()}")
tn.head()
```

Dropped 0 rows with a \$0 placeholder budget

Duplicate titles remaining (legitimate remakes/re-releases): 84

```
Out[7]:
```

	id	release_date	title	production_budget	domestic_gross	worldwide_gross	year
0	1	2009-12-18	Avatar	425000000.0	760507625.0	2.776345e+09	2009
1	2	2011-05-20	Pirates of the Caribbean: On Stranger Tides	410600000.0	241063875.0	1.045664e+09	2011
2	3	2019-06-07	Dark Phoenix	350000000.0	42762350.0	1.497624e+08	2019
3	4	2015-05-01	Avengers: Age of Ultron	330600000.0	459005868.0	1.403014e+09	2015
4	5	2017-12-15	Star Wars Ep. VIII: The Last Jedi	317000000.0	620181382.0	1.316722e+09	2017

Step 2: Clean Box Office Mojo (bom)

- Drop the 5 rows missing a `studio` label (cannot be used for the studio question regardless).
- Strip commas from `foreign_gross` and cast to `float` . We keep rows with a missing `foreign_gross` in the dataset (rather than dropping them outright) since they are still usable for the genre/financial-return question - we only require `foreign_gross` to be present when answering the international-audience question.

```
In [8]: bom = bom_raw.copy()
bom = bom.dropna(subset=["studio"])
bom["foreign_gross"] = bom["foreign_gross"].str.replace(",", "", regex=False).astype(float)

print(f"bom shape after cleaning: {bom.shape}")
print(f"Missing foreign_gross: {bom['foreign_gross'].isna().sum()} of {len(bom)} rows ({bom['foreign_gross'].head()})")
```

bom shape after cleaning: (3382, 5)
Missing foreign_gross: 1349 of 3382 rows (40%)

```
Out[8]:
```

	title	studio	domestic_gross	foreign_gross	year
0	Toy Story 3	BV	415000000.0	652000000.0	2010
1	Alice in Wonderland (2010)	BV	334200000.0	691300000.0	2010
2	Harry Potter and the Deathly Hallows Part 1	WB	296000000.0	664300000.0	2010
3	Inception	WB	292600000.0	535700000.0	2010
4	Shrek Forever After	P/DW	238700000.0	513900000.0	2010

Step 3: Clean IMDB (im)

- Inner-join `movie_basics` with `movie_ratings` on `movie_id` , which automatically restricts us to titles with enough audience votes to have a rating.
- Drop rows with a missing `genres` value, since genre is central to all three business questions.

```
In [9]: im = basics_raw.merge(ratings_raw, on="movie_id", how="inner")
im = im.rename(columns={"primary_title": "title", "start_year": "year"})

before = len(im)
im = im.dropna(subset=["genres"])
print(f"Dropped {before - len(im)} rows with missing genre data")
print(f"im shape after cleaning: {im.shape}")
im.head()
```

Dropped 804 rows with missing genre data
im shape after cleaning: (73052, 7)

Out[9]:	movie_id	title	year	runtime_minutes	genres	averagerating	numvotes
0	tt0063540	Sunghursh	2013	175.0	Action,Crime,Drama	7.0	77
		One Day					
1	tt0066787	Before the Rainy Season	2019	114.0	Biography,Drama	7.2	43
2	tt0069049	The Other Side of the Wind	2018	122.0	Drama	6.9	4517
3	tt0069204	Sabse Bada Sukh	2018	NaN	Comedy,Drama	6.1	13
4	tt0100275	The Wandering Soap Opera	2017	80.0	Comedy,Drama,Fantasy	6.5	119

Step 4: Merge all three sources into one analytical dataset

- Merge `tn` with `im` on matching `(title, year)`.
- Drop duplicate `movie_id` matches (a small number of titles can match more than one IMDB entry).
- Merge the result with `bom` on `(title, year)` to bring in `studio` and `foreign_gross`.
- Drop duplicates again after the second merge, for the same reason.

```
In [10]: df = tn.merge(im, on=["title", "year"], how="inner")
df = df.drop_duplicates(subset="movie_id")
print(f"tn + im merged: {df.shape}")

df = df.merge(bom[["title", "year", "studio", "foreign_gross"]], on=["title", "year"], how="inner")
df = df.drop_duplicates(subset="movie_id")
print(f"+ bom merged (final): {df.shape}")

df.isna().sum()

tn + im merged: (1496, 12)
+ bom merged (final): (1042, 14)
```

```
Out[10]: id                0
release_date             0
title                   0
production_budget        0
domestic_gross           0
worldwide_gross          0
year                    0
movie_id                0
runtime_minutes          2
genres                  0
averagerating           0
numvotes                0
studio                  0
foreign_gross           103
dtype: int64
```

Step 5: Engineer analysis columns

- `roi`: return on investment, as a multiple of budget - $(\text{worldwide_gross} - \text{production_budget}) / \text{production_budget}$
- `profit_musd`: profit in millions of USD
- `intl_share`: the share of a film's combined domestic + foreign gross that came from outside the U.S. - $\text{foreign_gross} / (\text{domestic_gross} + \text{foreign_gross})$. This is only computable for the ~89% of rows with a non-missing `foreign_gross`.

```
In [11]: df["roi"] = (df["worldwide_gross"] - df["production_budget"]) / df["production_budget"]
df["profit_musd"] = (df["worldwide_gross"] - df["production_budget"]) / 1_000_000
df["intl_share"] = df["foreign_gross"] / (df["domestic_gross"] + df["foreign_gross"])

print(f"Final analytical dataset: {df.shape[0]} films, {df.shape[1]} columns")
```

```
df[["title", "year", "studio", "production_budget", "worldwide_gross", "roi", "profit_musd", "intl_shr"]
```

Final analytical dataset: 1042 films, 17 columns

Out[11]:

	count	unique		top	freq	mean	std
title	1042	1025		Coco	2	NaN	NaN
year	1042.0	NaN		NaN	NaN	2013.651631	2.548387
studio	1042	78		Uni.	113	NaN	NaN
production_budget	1042.0	NaN		NaN	NaN	52373735.74856	58785097.912439
worldwide_gross	1042.0	NaN		NaN	NaN	173630642.555662	250309635.14046
roi	1042.0	NaN		NaN	NaN	3.500938	14.151249
profit_musd	1042.0	NaN		NaN	NaN	121.256907	207.870449
intl_share	939.0	NaN		NaN	NaN	0.505413	0.220491
genres	1042	180	Adventure,Animation,Comedy	60		NaN	NaN
averagerating	1042.0	NaN		NaN	NaN	6.464971	0.94289

Data cleaning / preparation notes and limitations:

- Matching on (title, year) is not perfect - a small number of legitimately re-titled or dual-year releases may be dropped, and two different films that happen to share both a title and release year could theoretically collide. Spot checks below confirm no unexpected duplicate movie_id values remain.
- foreign_gross is missing for roughly 10% of the merged dataset; we exclude those rows specifically for Question 2, rather than dropping them from the whole analysis, since they are still valid for Questions 1 and 3.
- ROI is heavily right-skewed by a handful of ultra-low-budget films with enormous payoffs (e.g. a horror film budgeted at \$100K that grossed \$40M+). Rather than removing these as "errors" - they are legitimate data points - we use the **median** (not the mean) throughout as our primary summary statistic, since it is much more robust to this kind of outlier.
- No missing values remain in the columns used for the three core analyses below.

```
In [12]: # Spot check: confirm no duplicate movie_id rows remain
assert df["movie_id"].duplicated().sum() == 0
print("No duplicate movie_id values - merge is clean.")
df.isna().sum()[["production_budget", "worldwide_gross", "genres", "studio", "foreign_gross"]]
```

No duplicate movie_id values - merge is clean.

```
Out[12]: production_budget    0
worldwide_gross             0
genres                     0
studio                     0
foreign_gross              103
dtype: int64
```

Data Analysis & Visualization

Question 1: Which genres deliver the best financial return?

Since a film can have multiple genres, we "explode" the comma-separated genres column so each movie contributes one row per genre it belongs to, then compute the **median** ROI and **median** profit per genre. We restrict this to genres with at least 15 films in our dataset so a single hit or flop doesn't distort a small sample.

```
In [13]: exploded = df.assign(genre=df["genres"].str.split(",").explode("genre")
exploded["genre"] = exploded["genre"].str.strip()
```

```

g1 = (
    exploded.groupby("genre")
    .agg(
        n_movies=("title", "count"),
        median_roi=("roi", "median"),
        median_profit_musd=("profit_musd", "median"),
    )
    .query("n_movies >= 15")
    .sort_values("median_roi", ascending=False)
)
TOP_GENRES = g1.head(5).index.tolist()
print("Top 5 genres by median ROI:", TOP_GENRES)
g1.round(2)

```

Top 5 genres by median ROI: ['Mystery', 'Horror', 'Sci-Fi', 'Animation', 'Thriller']

Out[13]:

	n_movies	median_roi	median_profit_musd
genre			
Mystery	81	3.45	55.99
Horror	107	2.90	42.28
Sci-Fi	96	2.23	167.05
Animation	84	2.23	237.78
Thriller	182	1.94	46.12
Music	31	1.92	25.28
Adventure	278	1.78	155.31
Comedy	387	1.67	50.33
Romance	141	1.66	22.34
Action	327	1.41	77.55
History	30	1.35	46.40
Biography	104	1.32	24.76
Drama	508	1.25	21.78
Fantasy	88	1.18	43.83
Crime	161	1.17	26.80
Family	66	1.08	51.93
Sport	21	0.82	28.53

```

In [14]: fig, ax = plt.subplots(figsize=(11, 7.5))
colors = [PALETTE[0] if g in TOP_GENRES[:3] else PALETTE[4] for g in g1.index]

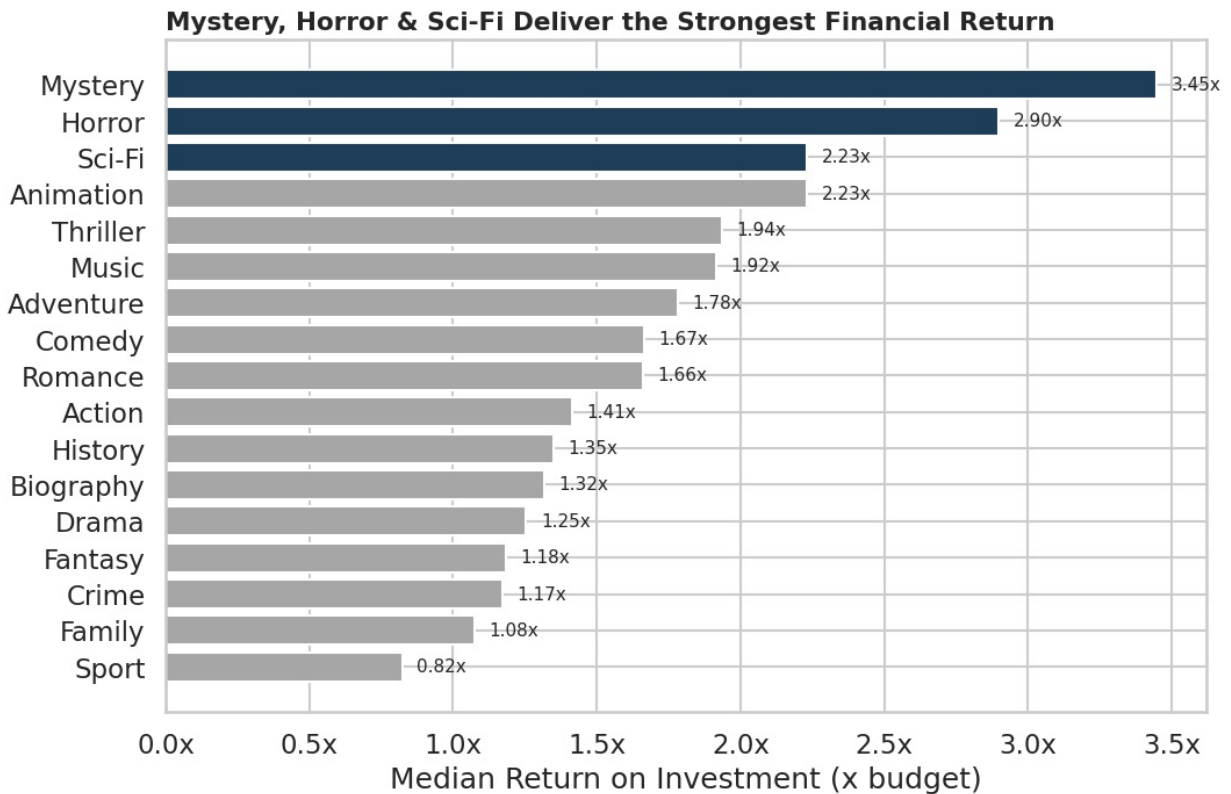
ax.barh(g1.index[::-1], g1["median_roi"][::-1], color=colors[::-1])
ax.axvline(0, color="black", linewidth=0.8)
ax.set_xlabel("Median Return on Investment (x budget)")
ax.set_title(
    "Mystery, Horror & Sci-Fi Deliver the Strongest Financial Return",
    fontsize=15.5, fontweight="bold", loc="left",
)
ax.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"{x:.1f}x"))
for i, v in enumerate(g1["median_roi"][::-1]):
    ax.text(v + 0.05, i, f"{v:.2f}x", va="center", fontsize=11)

plt.figtext(
    0.12, 0.01,
    f"Genres with fewer than 15 films excluded. n={df.shape[0]} films.",
    fontsize=9, color="gray",
)
plt.tight_layout(rect=[0, 0.03, 1, 1])
plt.savefig("images/viz1_genre_roi.png", dpi=200, bbox_inches="tight")

```



```
plt.show()
```



Genres with fewer than 15 films excluded. n=1042 films.

Finding 1: Mystery (median ROI 3.45x, median profit \$56.0M) and **Horror** (median ROI 2.90x, median profit \$42.3M) deliver by far the strongest financial return of any major genre, with **Sci-Fi** and **Animation** tied for third (2.23x ROI, though Animation's median profit of \$237.8M is dramatically higher thanks to bigger budgets and grosses). Genres like Drama, Fantasy, Crime, and Family cluster near or below the 1.0x-1.3x range, meaning a typical film in those genres barely breaks even or only modestly profits relative to its budget.

Question 2: Which genres do best with an international audience?

We measure international performance as `intl_share`: the percentage of a film's combined domestic + foreign box office that came from outside the U.S. A genre with a high median `intl_share` "travels well" internationally relative to its U.S. performance, regardless of the film's absolute size. We compare each genre's median share against the **all-genre median** to see which genres over- or under-index internationally.

```
In [15]: df_intl = df.dropna(subset=["foreign_gross"]).copy()
exploded_intl = df_intl.assign(genre=df_intl["genres"].str.split(",")).explode("genre")
exploded_intl["genre"] = exploded_intl["genre"].str.strip()

overall_median_share = df_intl["intl_share"].median()

g2 = (
    exploded_intl.groupby("genre")
    .agg(n_movies=("title", "count"), median_intl_share=("intl_share", "median"))
    .query("n_movies >= 15")
    .sort_values("median_intl_share", ascending=False)
)
print(f"All-genre median international share: {overall_median_share:.1%}")
g2.round(3)
```

All-genre median international share: 52.3%

Out[15]:

	n_movies	median_intl_share
Adventure	273	0.645
Fantasy	79	0.632
Action	311	0.622
Sci-Fi	95	0.621
Animation	83	0.612
History	27	0.575
Thriller	167	0.556
Horror	94	0.526
Mystery	79	0.513
Family	63	0.508
Biography	85	0.475
Comedy	350	0.470
Drama	424	0.469
Crime	148	0.457
Romance	117	0.441
Music	27	0.436
Sport	18	0.284

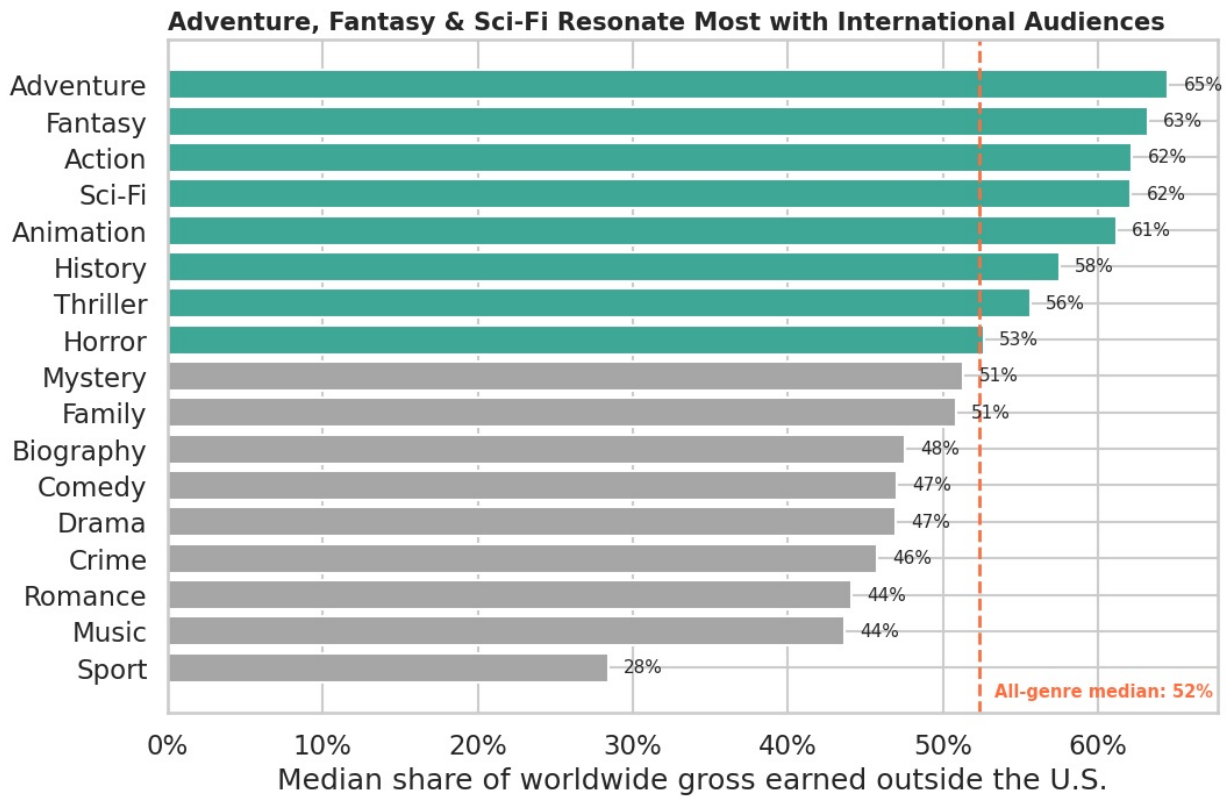
genre

Adventure	273	0.645
Fantasy	79	0.632
Action	311	0.622
Sci-Fi	95	0.621
Animation	83	0.612
History	27	0.575
Thriller	167	0.556
Horror	94	0.526
Mystery	79	0.513
Family	63	0.508
Biography	85	0.475
Comedy	350	0.470
Drama	424	0.469
Crime	148	0.457
Romance	117	0.441
Music	27	0.436
Sport	18	0.284

```
In [16]: fig, ax = plt.subplots(figsize=(11, 7.5))
colors2 = [PALETTE[3] if v > overall_median_share else PALETTE[4] for v in g2["median_intl_share"]]

ax.barh(g2.index[::-1], g2["median_intl_share"][::-1] * 100, color=colors2[::-1])
ax.axvline(overall_median_share * 100, color=PALETTE[1], linewidth=2, linestyle="--")
ax.text(
    overall_median_share * 100 + 1, -0.8,
    f"All-genre median: {overall_median_share:.0%}",
    color=PALETTE[1], fontsize=10.5, fontweight="bold",
)
ax.set_xlabel("Median share of worldwide gross earned outside the U.S.")
ax.set_title(
    "Adventure, Fantasy & Sci-Fi Resonate Most with International Audiences",
    fontsize=15, fontweight="bold", loc="left",
)
ax.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"{x:.0f}%"))
for i, v in enumerate(g2["median_intl_share"][::-1]):
    ax.text(v * 100 + 1, i, f"{v*100:.0f}%", va="center", fontsize=11)

plt.figtext(
    0.12, 0.01,
    f"Genres with fewer than 15 films excluded. Films missing foreign gross data excluded. n={df_intl",
    fontsize=9, color="gray",
)
plt.tight_layout(rect=[0, 0.03, 1, 1])
plt.savefig("images/viz2_genre_international.png", dpi=200, bbox_inches="tight")
plt.show()
```



Genres with fewer than 15 films excluded. Films missing foreign gross data excluded. n=939 films.

Finding 2: **Adventure** (65%), **Fantasy** (63%), **Action** (62%), **Sci-Fi** (62%), and **Animation** (61%) earn well over half their box office outside the U.S. - meaningfully above the all-genre median of 52%. Notably, this is a **different set of genres** than our top-ROI list from Question 1: **Horror** and **Mystery** (our best financial performers) sit right around the median (53% and 51%), meaning they perform reasonably well internationally but do not especially over-index there. **Sport** is a clear outlier at just 28% international share - it is an overwhelmingly domestic genre.

Question 3: Which studios perform best, and do they align with the top genres?

We compute median ROI per studio, restricted to studios with at least 8 films in our dataset (to avoid ranking a studio on 1-2 lucky or unlucky releases). For each studio, we also compute the **percentage of its catalog that falls within our top-5 ROI genres from Question 1** (Mystery, Horror, Sci-Fi, Animation, Thriller), and compare that against the market-average percentage, to see whether top-performing studios actually specialize in top-performing genres.

```
In [17]: studio_stats = (
    df.groupby("studio")
      .agg(n_movies=("title", "count"), median_roi=("roi", "median"))
      .query("n_movies >= 8")
      .sort_values("median_roi", ascending=False)
    )

# For each studio, what % of its catalog falls within the Q1 top-5 genres?
eligible = df[df["studio"].isin(studio_stats.index)].copy()
eligible["has_top_genre"] = eligible["genres"].apply(
    lambda g: any(tg in [x.strip() for x in g.split(",")] for tg in TOP_GENRES)
)
studio_stats["pct_top_genre_films"] = eligible.groupby("studio")["has_top_genre"].mean() * 100

market_avg_alignment = df["genres"].apply(
    lambda g: any(tg in [x.strip() for x in g.split(",")] for tg in TOP_GENRES)
)
```

```

).mean() * 100

print(f"Market-average share of films in the top-5 ROI genres: {market_avg_alignment:.1f}%")
studio_stats.round(2)

```

Market-average share of films in the top-5 ROI genres: 39.9%

Out[17]: **n_movies** **median_roi** **pct_top_genre_films**

studio			
WB (NL)	37	2.84	21.62
P/DW	10	2.63	80.00
Wein.	30	2.53	20.00
TriS	17	2.46	29.41
BV	71	2.21	43.66
Uni.	113	2.19	43.36
SGem	24	2.18	45.83
Sony	68	1.93	48.53
A24	13	1.79	30.77
Fox	103	1.77	46.60
Par.	66	1.76	54.55
CBS	10	1.59	40.00
LG/S	26	1.54	42.31
LGF	54	1.43	53.70
STX	14	1.40	21.43
ORF	24	1.38	37.50
Focus	33	1.38	36.36
SPC	19	1.28	10.53
WB	95	1.18	37.89
Rela.	23	1.04	52.17
FoxS	43	0.85	20.93
Sum.	12	0.63	33.33
RAtt.	15	0.50	40.00
Magn.	8	-0.60	37.50
IFC	8	-0.73	37.50

```

In [18]: fig, ax = plt.subplots(figsize=(11.5, 7.5))
          sizes = studio_stats["n_movies"] * 6
          top5 = studio_stats.sort_values("median_roi", ascending=False).head(5)
          colors3 = [PALETTE[0] if s in top5.index else PALETTE[4] for s in studio_stats.index]

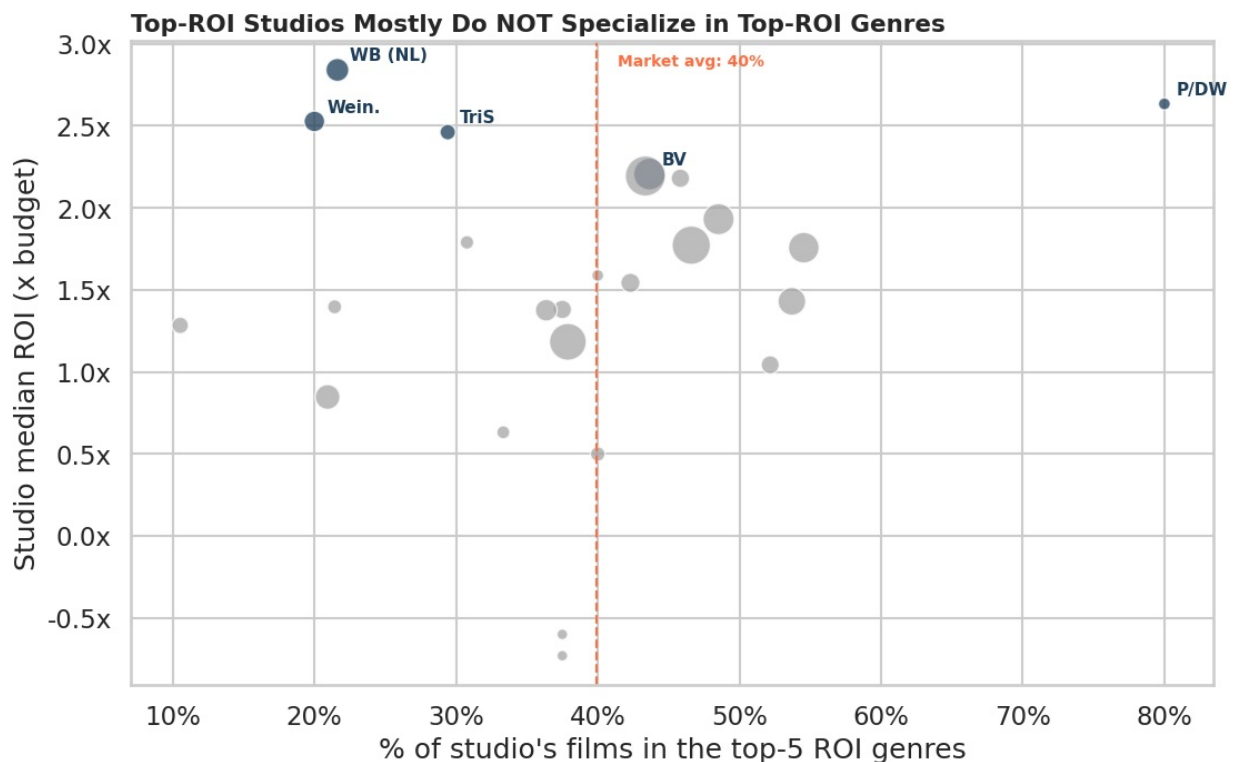
          ax.scatter(
              studio_stats["pct_top_genre_films"], studio_stats["median_roi"],
              s=sizes, c=colors3, alpha=0.75, edgecolor="white", linewidth=1,
          )
          for studio in top5.index:
              row = studio_stats.loc[studio]
              ax.annotate(
                  studio, (row["pct_top_genre_films"], row["median_roi"]),
                  textcoords="offset points", xytext=(8, 6),
                  fontsize=11, fontweight="bold", color=PALETTE[0],
              )

```

```

ax.axvline(market_avg_alignment, color=PALETTE[1], linestyle="--", linewidth=1.5)
ax.text(
    market_avg_alignment + 1.5, ax.get_ylim()[1] * 0.95,
    f"Market avg: {market_avg_alignment:.0f}%",
    color=PALETTE[1], fontsize=10, fontweight="bold",
)
ax.set_xlabel("% of studio's films in the top-5 ROI genres")
ax.set_ylabel("Studio median ROI (x budget)")
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"{x:.1f}x"))
ax.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"{x:.0f}%"))
ax.set_title(
    "Top-ROI Studios Mostly Do NOT Specialize in Top-ROI Genres",
    fontsize=15.5, fontweight="bold", loc="left",
)
plt.figtext(
    0.12, 0.01,
    f"Studios with fewer than 8 films excluded. Bubble size = number of films. Top 5 studios by median ROI",
    fontsize=9, color="gray",
)
plt.tight_layout(rect=[0, 0.03, 1, 1])
plt.savefig("images/viz3_studio_alignment.png", dpi=200, bbox_inches="tight")
plt.show()

```



Studios with fewer than 8 films excluded. Bubble size = number of films. Top 5 studios by median ROI labeled. n=1042 films.

Finding 3: The top 5 studios by median ROI are **New Line/WB (NL)** (2.84x), **Paramount/DreamWorks Animation (P/DW)** (2.63x), **The Weinstein Company (Wein.)** (2.53x), **TriStar (TriS)** (2.46x), and **Buena Vista/Disney (BV)** (2.21x). But when we check what share of each studio's catalog falls into our top-5 ROI genres (39.9% is the market average), the pattern is surprising: **only P/DW (80%) and BV (44%) exceed the market average** - New Line, Weinstein, and TriStar all sit *below* the market average (22%, 20%, and 29% respectively) despite having the best studio-level ROI in the dataset. In other words, **strong studio performance and genre choice are only loosely connected** - three of our five best-performing studios succeeded largely *outside* the top-ROI genres, suggesting execution, talent, and marketing matter as much as (or more than) genre selection alone.

Conclusions & Recommendations

Based on the three analyses above, we recommend the new movie studio adopt the following guidelines:

1. Prioritize Mystery and Horror for the strongest financial return

Mystery (median ROI 3.45x, median profit \$56.0M) and Horror (median ROI 2.90x, median profit \$42.3M) are the clear financial leaders. Both also carry modest average budgets relative to their return, making them a lower-risk way to build an early track record.

2. Look to Adventure, Fantasy, Sci-Fi, and Animation for international growth

If the studio's ambitions include building a global audience (or a franchise with international licensing potential), these genres earn 60%+ of their gross outside the U.S. - well above the 52% all-genre median. Note the trade-off: these genres did not top our ROI list in Question 1, so this is a different strategic lever (audience reach) rather than pure financial efficiency.

3. Study execution at New Line, Weinstein, and TriStar - not just their genre mix

The studios with the best ROI often succeeded *outside* the top-ROI genres, which means their success is not simply explained by "picking the right genre." The new studio should study what these studios did differently in production, marketing, and release strategy, rather than assuming genre selection alone drives studio-level performance.

Taken together: our recommended approach is to build an early slate anchored in Mystery and Horror for reliable financial return, while treating genres like Adventure and Sci-Fi as a longer-term, higher-budget bet for international reach - and to benchmark operationally against New Line and Weinstein's playbook rather than assuming their genre mix explains their success.

Limitations & Next Steps

- This analysis is correlational, not causal - we cannot prove that a genre or studio *causes* higher returns, only that the two are historically associated. Marketing spend, release timing, star power, and franchise/IP status are confounding variables not captured here.
- `foreign_gross` was missing for about 10% of the merged dataset, slightly reducing the sample available for Question 2.
- Rotten Tomatoes and TheMovieDB data were excluded due to a lack of a clean join key to the other sources; incorporating critic scores could add a quality dimension to this analysis.
- A natural next step is to build these findings out as an interactive Tableau dashboard (see the accompanying Tableau workbook) so studio leadership can filter by genre and studio themselves, and to layer in release timing and marketing spend as additional levers.